The documentation and process conversion measures necessary to comply with this revision shall be completed by 6 June 1998.

INCH-POUND

MIL-PRF-19500/495B
6 March 1998
SUPERSEDING
MIL-S-19500/495A(USAF)
28 May 1993

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, UNITIZED, DUAL-TRANSISTOR, NPN, SILICON, TYPES 2N5793, 2N5794 AND 2N5794U, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for two electrically isolated, unmatched, NPN, silicon, transistors as one dual unit . Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figures 1 (similar to T0-99) and 2 (surface mount).

1.3 Maximum ratings. T_A = +25°C, unless otherwise specified.

P_T 1/ T_A = +25°C		I_C	V_{CBO}	V_{CEO}	V_{EBO}	T_{OP} and T_{STG}
One section	Total device					
W	W	mA dc	V dc	V dc	V dc	°C
0.5	0.6	600	75	40	6.0	-65 to +175

1/ For $T_A \geq$ 25°C, derate linearly 2.86 mW/°C one section, 3.43 mW/°C total.

	C_{obo}	$\|h_{fe}\|$	Switching	
	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	V_{CE} = 20 V dc I_C = 20 mA dc f = 100 MHz	t_{on}	t_{off}
	pF		ns	ns
Minimum		2.0		
Maximum	8.0	10.0	45	310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.

1.4 Primary electrical characteristics. T_A = +25° C, unless otherwise specified.

	h_{FE1}		h_{FE4} 1/		$V_{CE(sat)1}$ 1/	$V_{CE(sat)2}$ 1/	$V_{BE(sat)1}$ 1/
Limits	V_{CE} = 10 V dc I_C = 100 µA dc		V_{CE} = 10 V dc I_C = 150 mA dc		I_C = 150 mA dc I_B = 15 mA dc	I_C = 300 mA dc I_B = 30 mA dc	I_C = 150 mA dc I_B = 15 mA dc
	Min	Max	Min	Max			
					V dc	V dc	V dc
2N5793	20		40	120			0.6
2N5794	35		100	300	0.3	0.9	1.2
2N5794U	35		100	300	0.3	0.9	1.2

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARDS

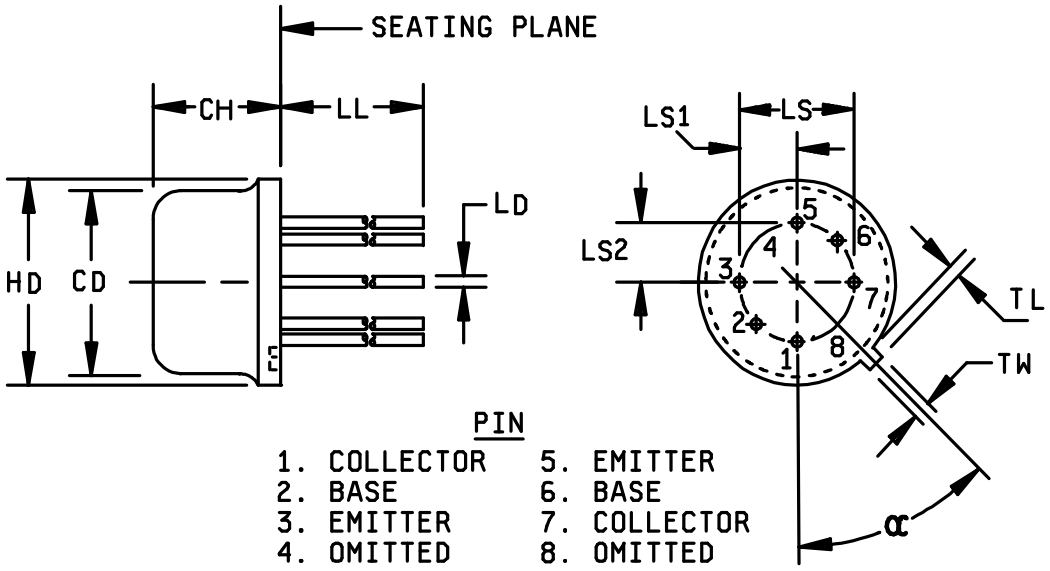
MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.

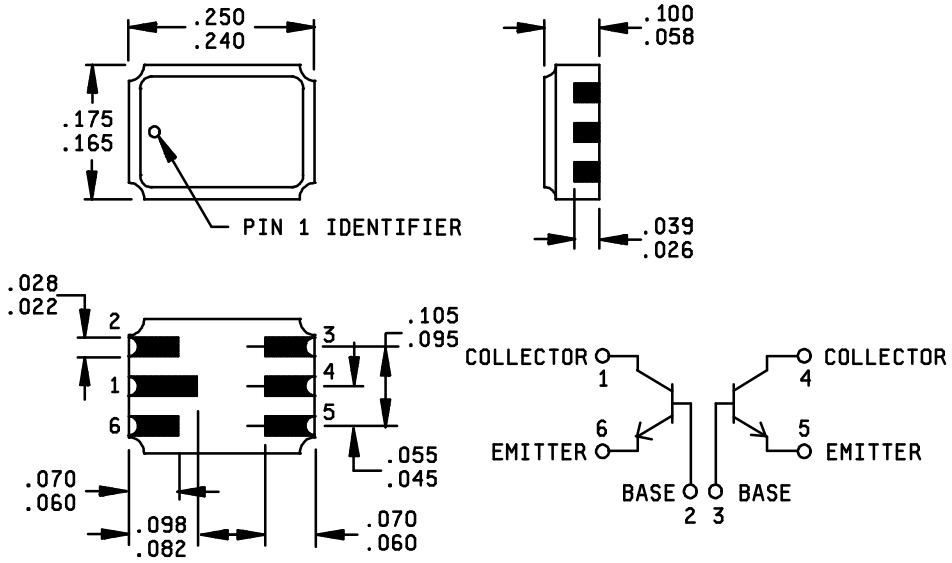


Dimensions					
	Inches		Millimeters		
Symbol	Min	Max	Min	Max	Note
CD	.305	.335	7.75	8.51	
CH	.150	.185	3.81	4.70	
HD	.335	.370	8.51	9.40	
LD	.016	.021	0.41	0.53	
LL	.500		12.70		
LS	.200 BSC		5.08 BSC		2
LS1	.100 BSC		2.54 BSC		
LS2	.100 BSC		2.54 BSC		
TL	.029	.045	0.74	1.14	1
TW	.028	.034	0.71	1.14	
α	45° TP		45° TP		4

NOTES:
1. Measured from maximum diameter of the product.
2. Leads having maximum diameter .019 inch (.483 mm) measured in gaging plan .054 inch (1.37 mm) + .001 inch (.025 mm) - .000 inch (.000 mm) below the seating plane of the product shall be within .007 inch (.178 mm) of their true position relative to a maximum width tab.
3. The product may be measured by direct methods or by gauge.
4. Tab centerline.

FIGURE 1. Physical dimensions (2N5793 and 2N5794).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.



Inches	mm
.022	0.56
.026	0.66
.028	0.71
.039	0.99
.045	1.14
.055	1.40
.058	1.47
.060	1.52
.066	1.68
.070	1.78

Inches	mm
.080	2.03
.082	2.08
.095	2.41
.098	2.49
.100	2.54
.105	2.67
.165	4.19
.175	4.44
.240	6.10
.250	6.35

FIGURE 2. Physical Dimensions (2N5794U).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1 and 2 (surface mount) herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, the marking of the country of origin may be omitted from the body of the device, but shall be retained on the initial container.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement
	JANTX and JANTXV levels
7	1/
9	not applicable
10	48 hours minimum
11	I_{CBO2} and h_{FE4}
12	Burn-in (see 4.3.1) 80 hours minimum
13	Subgroup 2 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc; whichever is greater. Δh_{FE4} = 15 percent of initial value.

1/ Hermetic seal test shall be performed in either screen 7 or screen 14.

4.3.1 Burn-in conditions. Burn-in conditions are as follows:

T_A = +30° C ±5° C; V_{CB} = 30 V dc; P_T = 300 mW each section (600 mW total device).

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed per E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied per 4.4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with 4.4.2.2 for JAN, JANTX, and JANTXV group B testing herein. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein (when specified below).

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1 2/	1039	HTRB, test condition A, T_A = 150° C, V_{CB} = 50 V dc, 340 hours, n = 22 devices, c = 0.
2 2/	1039	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 10 V dc, P_T = 300 mW (P_T = 600 mW total device); at T_A = room ambient per MIL-STD-750. No heat sink or forced-air cooling on the devices
	or	shall be permitted. n = 45 devices, Acc = 0
	1037	Intermittent operation life: V_{CB} = 10 V dc , 6,000 cycles, $\Delta T_J \geq +100°$ C; forced air cooling allowed on cooling cycle only. n = 45 devices, Acc = 0.
3	2037	Bond strength, n = 11 wires, c = 0.
4 2/, 3/	1032	High-Temperature life (non-operating), T_A = +200° C n = 22, c = 0
4 3/	1018	n = 3, c = 0

1/ Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

2/ Electrical endpoints for this step are table 1, group A, subgroup 2 herein.

3/ Not required for eutectic and solder die attach.

4.4.2.3 Group B sample selection. Samples selected from group B inspection (JAN, JANTX, and JANTXV) lots shall have samples selected from each wafer lot from a die fabrication line that the manufacturer chooses to "certify" for the die design used in products manufactured to this specification.

4.4.2.4 Product acceptable for delivery. The product shall be acceptable for delivery only after the successful completion of all group B testing and shall be comprised of die manufactured on the same certified die fabrication line as the sample selected for 4.4.2.3.

4.4.3 Group C inspection, JAN, JANTX, and JANTXV. Group C inspection (JAN, JANTX, and JANTXV, package related tests) shall be as follows and shall include those package or case-related tests which are performed periodically. Group C test on each package family shall be performed on devices from each six months production of devices (based on inspection lot identification codes) for each assembly location and die attach method. Electrical measurements (end-points) shall be after each step below and shall be those tests specified in table I, group A, subgroup 2 herein. The sample size for each of these subgroups is: 22 devices, c = 0. Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

Step	Method	Condition
1	2066	
2	1051	Test condition C.
3	2036	Test condition E; not applicable for U suffix devices.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 75 V dc,	I_{CBO}		10	µA dc
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	40		V dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6 V dc,	I_{EBO}		10	µA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO2}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO2}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 100 µA dc	h_{FE1}			
2N5793				20		
2N5794, 2N5794U				35		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}			
2N5793				25		
2N5794, 2N5794U				50		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE3}			
2N5793				35		
2N5794, 2N5794U				75		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE4}			
2N5793				40	120	
2N5794, 2N5794U				100	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 300 mA dc; pulsed (see 4.5.1)	h_{FE5}			
2N5793				25		
2N5794, 2N5794U				40		

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/	Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE6}			
2N5793				20		
2N5794, 2N5794U				50		
Collector to emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.3	V dc
Collector to emitter saturation voltage	3071	I_C = 300 mA dc; I_B = 30 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.9	V dc
Base to emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$	0.6	1.2	V dc
Base to emitter saturation voltage	3066	Test condition A; I_C = 300 mA dc; I_B = 30 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		1.8	V dc
Subgroup 3						
High temperature operation:		T_A = +150° C				
Collector to base cutoff current	3041	Bias condition D; V_{CB} = 50 V dc	I_{CBO3}		10	µA dc
Low temperature operation:		T_A = -55° C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE7}			
2N5793				16		
2N5794, 2N5794U				40		
Subgroup 4						
Magnitude of small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 20 V dc; I_C = 20 mA dc; f = 100 MHz	$\lvert h_{fe} \rvert$	2	10	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF

See footnote at end of table.

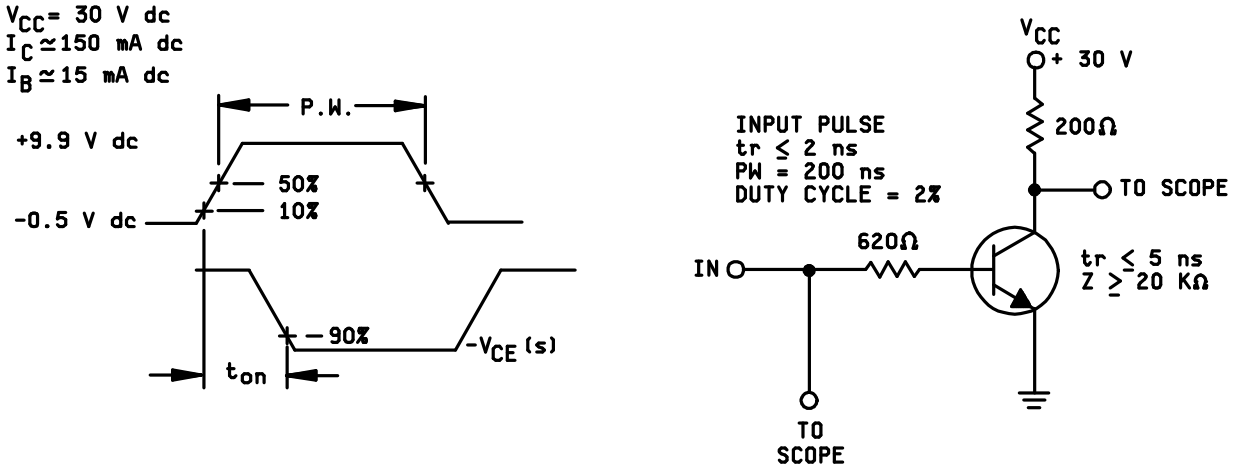
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

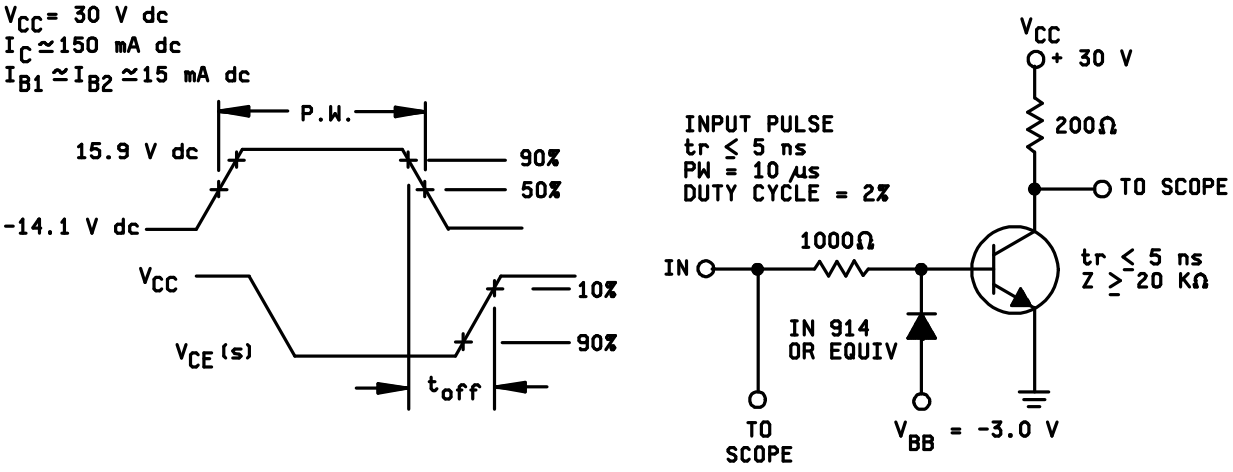
Inspection 1/	Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
Subgroup 4 - Continued						
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		33	pF
Pulse response:	3251	Test condition A, (see figure 3)				
Turn-on time		V_{CC} = 30 V dc; I_C = 150 mA dc; I_{B1} = 15 mA dc; $V_{BE(OFF)}$ = 0.5 V dc	t_{on}		45	ns
Turn-off time		V_{CC} = 30 V dc; I_C = 150 mA dc; I_{B1} = I_{B2} = 15 ma dc	t_{off}		310	ns
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.



TURN-ON (t_{on}) TIME TEST CIRCUIT



TURN-OFF (t_{off}) TIME TEST CIRCUIT

FIGURE 3. Switching time test circuits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1).

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

d. Packaging requirements (see 5.1).

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

CONCLUDING MATERIAL

Custodians:
Air Force - 17

Review activities
Air Force - 13, 19, 85

Preparing activity:
DLA - CC

(Project 5961-F150)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/495B	2. DOCUMENT DATE 6 March 1998

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, UNITIZED, DUAL-TRANSISTOR, NPN, SILICON, TYPES 2N5793, 2N5794 AND 2N5794U, JAN, JANTX, AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:27Z
Check the source to verify that this is the current version before use.